

Perikymata Distribution from the Final Curve

Given the final outer enamel surface (OES) $\mathbf{r}_f(s) = (x_f(s), y_f(s))$ and the termination time field $t^*(s)$, the perikymata distribution measures how much developmental time is compressed into each spatial region of the OES. Perikymata are essential markers of enamel termination time, such that 1 perikymata corresponds to $\mathcal{R}$ days of development, where $\mathcal{R}$ is the Retzius periodicity. The following steps outline how to compute the perikymata distribution along the OES. In other words, 1 PK corresponds to the distance the termination front progression in $\mathcal{R}$ days.

1 Establish a Reference Axis

In general, the reference axis is specified as an arbitrary unit vector $\hat{\mathbf{d}} \in \mathbb{R}^2$ and an origin point $\mathbf{r}_0$ on the OES. The projected coordinate of each point along the crown is then

$$\rho(s) = (\mathbf{r}_f(s) - \mathbf{r}_0) \cdot \hat{\mathbf{d}}. \quad (1)$$

The integration bounds are $\rho \in [0, H]$ where $H = \max_s \rho(s)$. When the geometry admits well-defined anatomical landmarks, $\hat{\mathbf{d}}$ and $\mathbf{r}_0$ can be set naturally from the cusp and cervix; in the general case they are free parameters that define the measurement orientation.

2 Partition the OES into Intervals

Divide $[0, H]$ into N equal intervals of width $\Delta = H/N$. Define a smooth indicator for the k -th interval:

$$\chi_k(s) = \sigma(\beta(\rho(s) - k\Delta)) \cdot \sigma(\beta((k+1)\Delta - \rho(s))), \quad (2)$$

where σ is the logistic function and β controls sharpness.

3 Integrate Developmental Time per Interval

$$\mathcal{T}_k = \int_0^L \left| \frac{dt^*}{ds} \right| \chi_k(s) ds \approx \sum_i dt_i^* \cdot \chi_k(s_{i+1/2}), \quad (3)$$

where $dt_i^* = |t^*(s_{i+1}) - t^*(s_i)|$.

4 Convert to Perikymata Count

Given Retzius periodicity $\mathcal{R}$ (days):

$$P_k = \frac{365 \mathcal{T}_k}{\mathcal{R}}. \quad (4)$$

Non-uniform $t^*(s)$ directly produces non-uniform P_k , since regions where the off-front decelerates accumulate more dt^* per unit projected length.

5 True PK distribution

In this regime, calculate number of perikymata as a function of distance along the arc length S of the OES (as opposed to the s arclength we get from the EDJ). Ultimately, one of the things we want to understand is how $S = S(s)$, that is the mapping from EDJ arclength to OES arclength.